

Ensemble-Gated SpO2 Correction

From an encoder-decoder-pretrained residual estimator that could not reliably beat a classical physics baseline, to a leakage-free ensemble-disagreement monitor that does. Full arc: architecture, two real bugs found and fixed, four rounds of LOSO tuning, a null result on single-model direction prediction, and a positive, held-out-validated closing-the-loop result.

Files train1.py → train2.py → train3.py **Dataset** 11 subjects • RValues_RawPPG_Aref_

Method 11-fold leave-one-subject-out **Date** 2026-09-03

EXECUTIVE SUMMARY

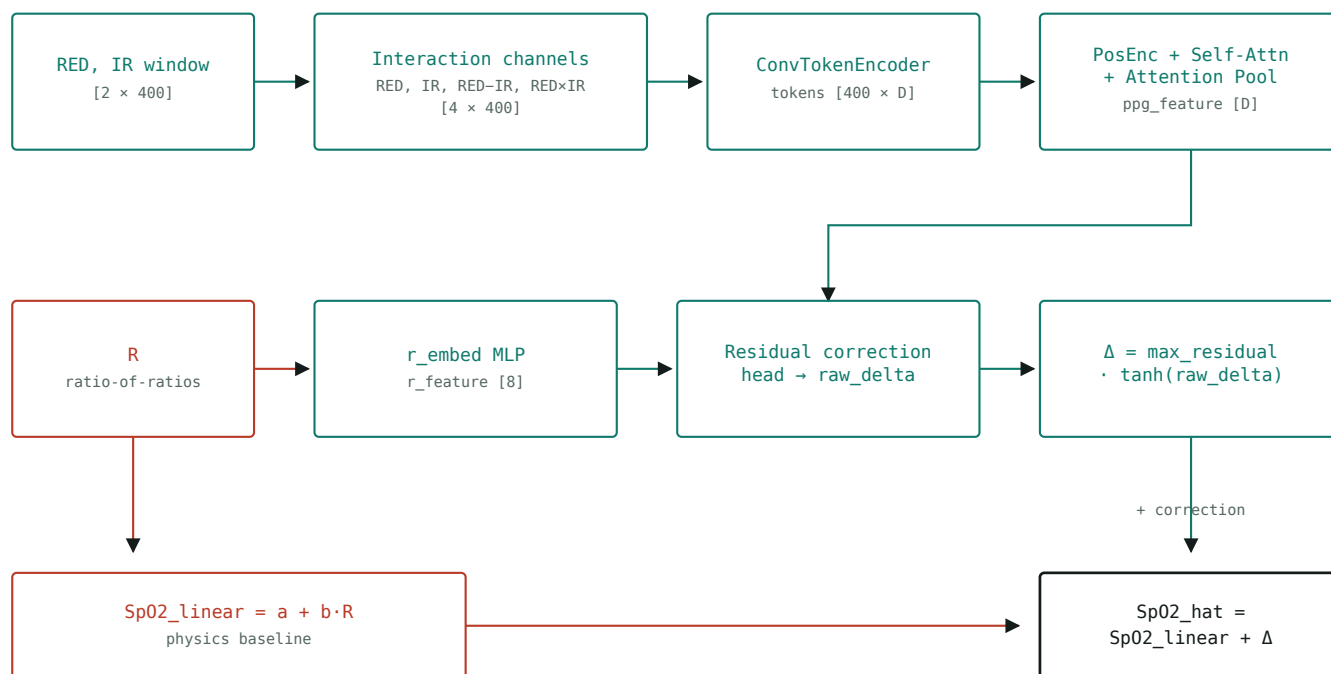
A physics-guided residual SpO2 estimator was pretrained with a self-supervised encoder-decoder, frozen, and fine-tuned in three stages. Two real implementation bugs were found and fixed along the way (a BatchNorm freeze leak, and a channel-variance bug that broke Stage-1 pretraining outright). Once fixed, rigorous 11-fold leave-one-subject-out (LOSO) evaluation showed the learned correction **did not reliably beat the classical ratio-of-ratios baseline** even after four rounds of targeted tuning — the best configuration landed within 0.8% of the baseline's RMSE, winning on 6/11 subjects. Diagnosing why revealed that the correction's own magnitude signal (ρ) predicts how much a correction helps or hurts *given* its direction is right or wrong, but not the direction itself — and no scalar a single trained model produces at inference (its raw pre-tanh logit, its internal embeddings, MC-dropout variance) predicts direction either. What does: **disagreement across a small, leakage-free ensemble of the already-trained LOSO models** (LOSO AUC 0.676, stable). Closing the loop — gating the residual's authority by that disagreement — produces a genuine, held-out-honest improvement: pooled soft-attenuation beats both the physics baseline and the uncontrolled hybrid model on MAE and RMSE, and on the single subject where the uncontrolled correction did the most damage, the gated version beats the physics baseline outright.

A physics baseline the network is only allowed to nudge

Every window is 400 samples (16 s at 25 Hz) of synchronized RED and IR PPG, paired with a ratio-of-ratios value R and a reference `SpO2_Rad` label. The model starts from the classical pulse-oximetry formula $\text{SpO2_linear} = a + b \cdot R$, fit once per fold by least squares on training subjects only, and learns a small, bounded correction on top of it:

```
SpO2_hat = SpO2_linear + max_residual · tanh(correction_head(ppg_feature,
r_feature))
```

The correction head's last layer is zero-initialized, so training starts exactly at the physics baseline and can only earn deviation from it through gradient descent.



Forward pass. Teal marks the learned path; red marks the fixed physics path. The two meet only at the final sum.

Three training stages share one encoder

The encoder is pretrained on raw signal reconstruction first (self-supervised, no SpO2 labels), then transplanted into the full model and frozen — genuinely frozen, including its BatchNorm

statistics, not just its gradients — then finally unfrozen for a gentler end-to-end fine-tune at a tenth of the head's learning rate.

1

Pretrain: MSE reconstruction of RED,
IR · lr 1e-3 · 150 epochs

2

Frozen head: Huber loss vs. SpO2,
encoder frozen · 60 epochs

3

Fine-tune: end-to-end, encoder lr 1e-4 /
head lr 5e-4 · 60 epochs

Two real bugs, found and fixed before any result could be trusted

Bug 1 • Freezing didn't actually freeze

The first pretrained-encoder pattern set `requires_grad = False` on the encoder and called it frozen. It wasn't: `ConvTokenEncoder` uses `BatchNorm1d`, whose running statistics are buffers, not parameters — they keep drifting under `model.train()` regardless of `requires_grad`, because `BatchNorm` normalizes with per-batch statistics whenever a module is in train mode. The fix: `set_encoder_trainable()` also calls `.train(trainable)` on the encoder, and a `train()` override re-pins it to `.eval()` even when the parent model is switched back to train mode. This exact fix was carried forward into every subsequent model in this project, including the MC-dropout evaluation in Section 6.

Bug 2 • The reconstruction loss exploded

The first Stage-1 pretraining run produced validation MSE in the **600,000–700,000** range and never converged — worse than the ~302 MSE of just predicting each channel's mean. Cause: the engineered `RED×IR` channel has roughly **300–600×** the variance of RED, IR, and RED-IR (measured on real data: 124,371 vs. 184/419/88). The decoder was asked to reconstruct all four channels under one unweighted MSE, so nearly all the loss and gradient came from one derived, redundant channel. Fix: keep the encoder's input at 4 channels, but have the decoder reconstruct only the 2 true underlying signals. Validation MSE then converged normally to ~65–70, roughly 4.5× better than the mean-prediction baseline.

WHY THIS MATTERS FOR EVERYTHING DOWNSTREAM

Both bugs would have silently produced a working-looking pipeline with a badly undertrained encoder or a leaking freeze. Every LOSO number in this report was generated only after both were fixed and independently verified against real training-run logs, not assumed.

Why a single train/val/test split lied, and what four LOSO rounds found

With both bugs fixed, one deterministic split (val=Subject11 , test=Subject2) looked like a clear win: fine-tuned validation RMSE **4.56** beat the physics baseline's **5.19**. But on the truly held-out test subject, fine-tuned RMSE was **6.93** — worse than the baseline's **6.25**. With only 11 subjects, one validation subject is a high-variance signal that can reward a correction fitting that one subject without generalizing. Every result after this point uses full 11-fold leave-one-subject-out (LOSO) evaluation instead, re-pretraining Stage 1 from scratch per fold so the held-out subject's raw signal never touches the encoder, labeled or not.

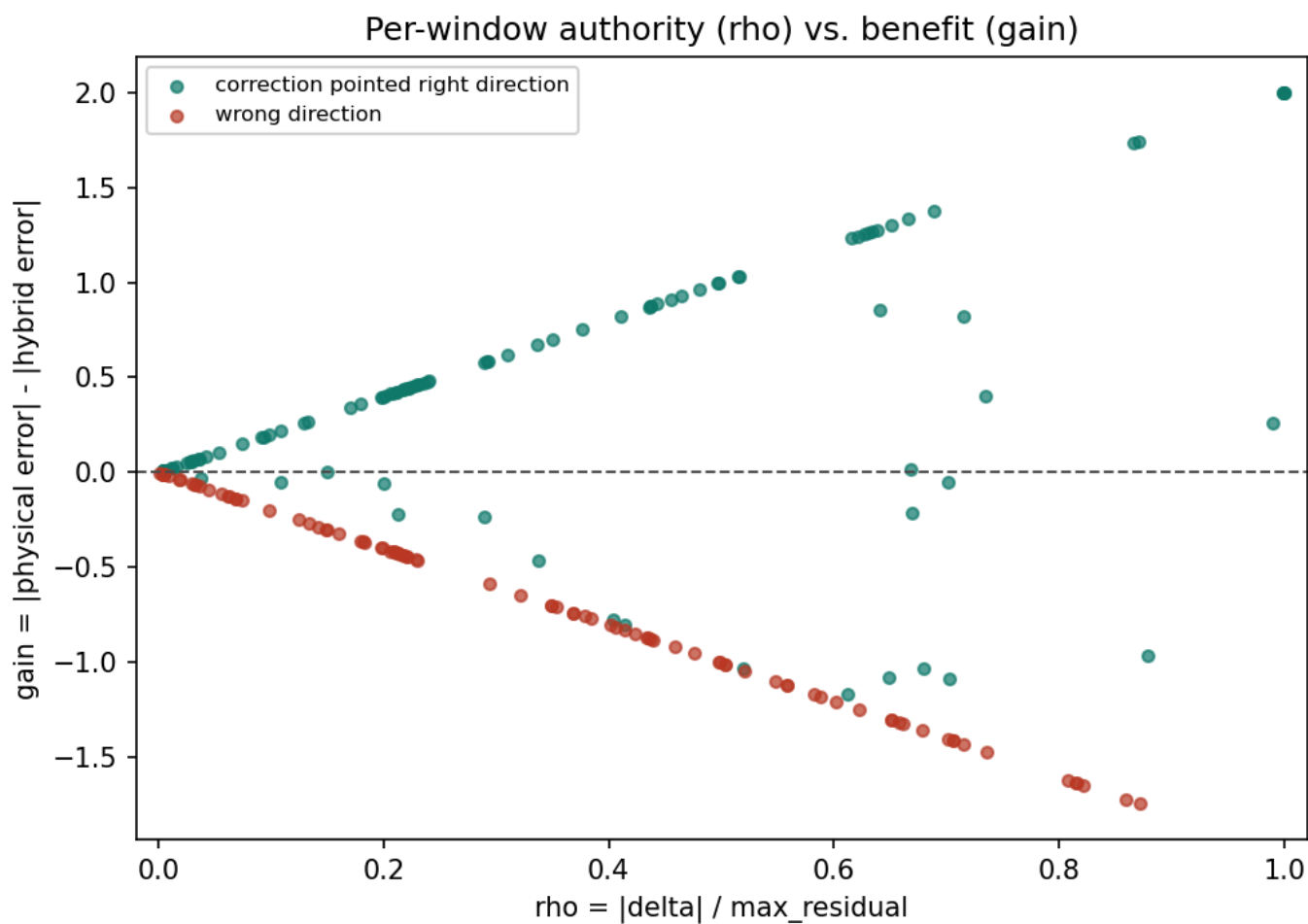
FOUR LOSO TUNING ROUNDS • MEAN TEST RMSE ACROSS 11 SUBJECTS

ROUND	CHANGE	LINEAR	FINE-TUNED	WINS
1	Baseline: hidden_dim=32, 1 val subject	3.191	3.295	4 / 11
2	Tighter output: max_residual 5 → 2, λ 0.001 → 0.01	3.191	3.281	5 / 11
3	Smaller model: hidden_dim 32 → 16, heads 4 → 2	3.191	3.216	6 / 11
4	2 validation subjects instead of 1	3.047	3.141	6 / 11

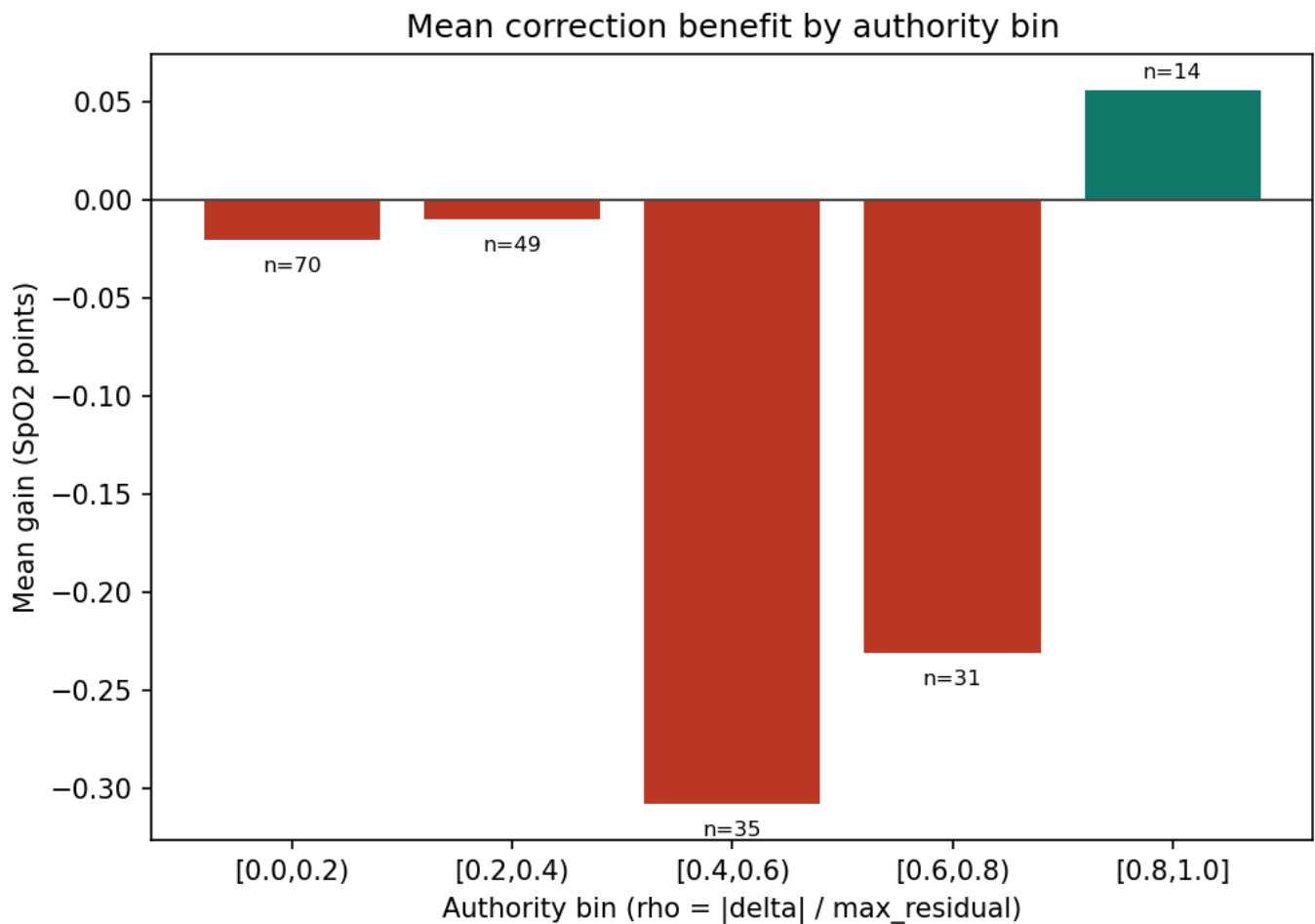
Round 3's capacity cut moved the gap roughly 3× more than Round 2's output regularization — the model had more parameters (34,842) than ~200 training windows could constrain; shrinking it to 14,394 was the bigger lever. Round 4 tested a specific hypothesis (one validation subject is a noisy early-stopping signal) and confirmed it for one fold while introducing two new regressions elsewhere — a plateau across two independent fixes, which is the signature of a data ceiling, not a remaining bug. **The best configuration found still does not beat the physics baseline in aggregate** — that honest negative result is the starting point for everything that follows.

ρ is a magnitude signal, not a reliability signal

Define, per window, $\text{gain} = |\text{physical error}| - |\text{hybrid error}|$ (positive = the correction helped) and $\rho = |\Delta| / \text{max_residual}$ (how hard the correction is pushing, 0–1). A binned mean-gain table looks non-monotonic and almost reassuring — the highest-authority bin $[0.8, 1.0]$ averages a slightly *positive* gain. The scatter plot below shows why that average is misleading:



Two nearly-perfect linear fans, not noise: right-direction corrections (teal) gain almost linearly with ρ up to +2.0; wrong-direction corrections (red) lose almost as linearly, down to -1.7. Pooled correlation is weak (Pearson $r = -0.065$, n.s.) only because both fans are averaged together.



The binned mean, read alone, would suggest [0.8,1.0] is the *safest* bin — an artifact of which subjects happen to dominate that bin's direction mix, not evidence that high authority is safe.

Conclusion: a runtime monitor that shrinks authority purely as a function of ρ would suppress the most helpful corrections along with the most harmful ones — ρ alone cannot tell them apart. The next two sections ask what can.

Can a single model's own output predict its direction? No.

Offline label (analysis only, never used at runtime): `direction_correct = sign( $\Delta$ ) == sign(y_true - y_linear)`. Nine candidate predictors were tested — ρ , $|\Delta|$, `y_linear`, `y_hat`, signed Δ , and four subject-independent combinations — each scored with the full LOSO discipline: pooled AUC/PR-AUC, an 11× leave-one-subject-excluded AUC stability check, and a threshold selected on 10 subjects and scored only on the 11th.

CSV-ONLY CANDIDATES • RANKED BY LOSO-STABLE AUC

CANDIDATE	AUC	LOSO AUC	HOLDOUT ACC.
delta (signed)	0.599	0.598±0.028	47.2%
delta × y_linear	0.598	0.598±0.027	53.8%
y_hat (LOSO z-score)	0.569	0.569±0.022	47.2%
y_linear	0.552	0.552±0.024	49.2%
abs(delta) / rho (rank-identical)	0.503	0.503±0.018	41.2%

Majority-class baseline was 53.3% accuracy. The best candidate (signed `delta`, LOSO AUC 0.598) scored a held-out threshold accuracy of only 47.2% — *worse than doing nothing*, despite an AUC nominally above chance. That's the report's clearest lesson in the difference between a ranking metric and a deployable rule: **none of the nine candidates is a usable monitor**. All are post-hoc scalar summaries of a single model's already-committed output; none is that model's internal state at the moment it decided direction.

What a single model can't tell you, an ensemble can

Extending the inference-only collection (no retraining) to save what `forward()` already computes and discards — the pre-tanh correction logit, the pooled attention embeddings, MC-dropout variance (with BatchNorm correctly pinned in eval mode) — plus one genuinely new signal: disagreement across a **leakage-free 3-model "clean ensemble"** per subject (that subject's own held-out fold model, plus the exactly two other fold models whose training set also excluded it, since `multi_subject_split` excludes both val and test subjects from training).

ENRICHED RUNTIME SIGNALS • RANKED BY LOSO-STABLE AUC

CANDIDATE	AUC	LOSO AUC	HOLDOUT ACC.
cross_model_std_delta	0.677	0.676±0.013	60.8%
cross_model_sign_agreement	0.622	0.622±0.025	62.8%
ppg_feature_norm	0.610	0.610±0.023	47.7%
raw_delta (pre-tanh)	0.599	0.598±0.028	47.2%
mc_dropout_delta_std	0.508	0.509±0.017	44.2%

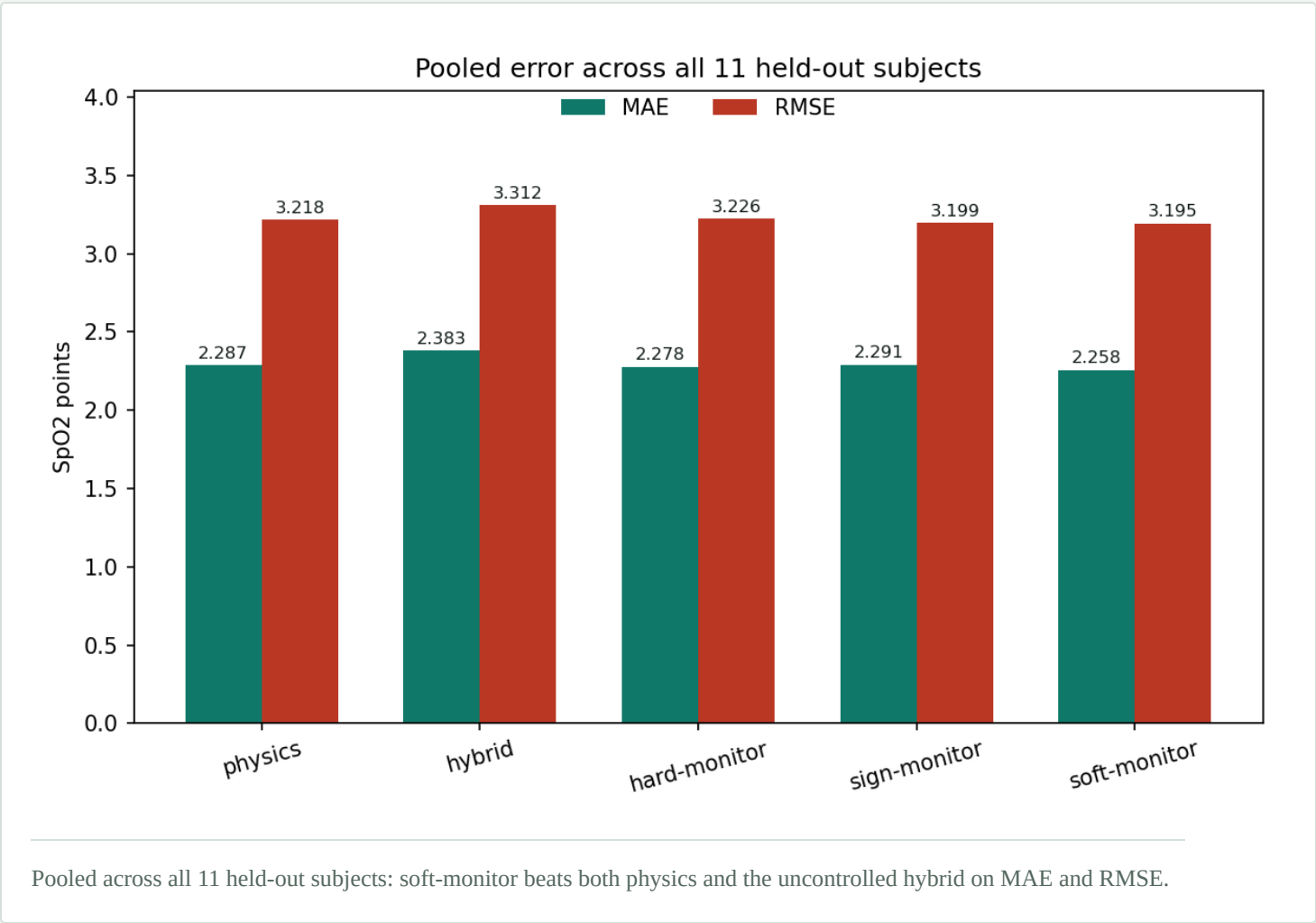
`cross_model_std_delta` is the first candidate in the whole investigation to clear both bars: LOSO AUC 0.676 (stable, range [0.655, 0.706] across 11 leave-one-subject-excluded recomputations), 60.8% pooled held-out threshold accuracy vs. 53.3% baseline, and — the metric that matters most — **accuracy across held-out subjects (mean ± spread) of 60.2% ± 13.2%, range [35.3%, 77.8%]**. Real, but uneven: it helps on most subjects and is worse than chance on the worst one.

A LOSO-honest logistic combination of the top 3 signals scored AUC 0.586 — *worse* than `cross_model_std_delta` alone (0.676), reported as-is rather than discarded: with ~180 training windows per fold, a fitted combination didn't beat the single best raw signal.

Closing the loop: does the signal actually improve the estimate?

Three monitored estimators, each with hyperparameters selected by a grid search on the *other* 10 subjects only, then frozen and applied to the held-out subject:

- **Hard monitor** — apply the full correction when `cross_model_std_delta` $\leq \tau$, otherwise fall back to physics.
- **Sign-agreement monitor** — apply the correction fully when the clean ensemble is unanimous on its sign; attenuate by $\beta \in \{0, 0.25, 0.5, 0.75\}$ otherwise, β chosen per fold.
- **Soft monitor** — continuous attenuation, `scale = clip(1 - std/ $\tau$ , 0, 1)`.

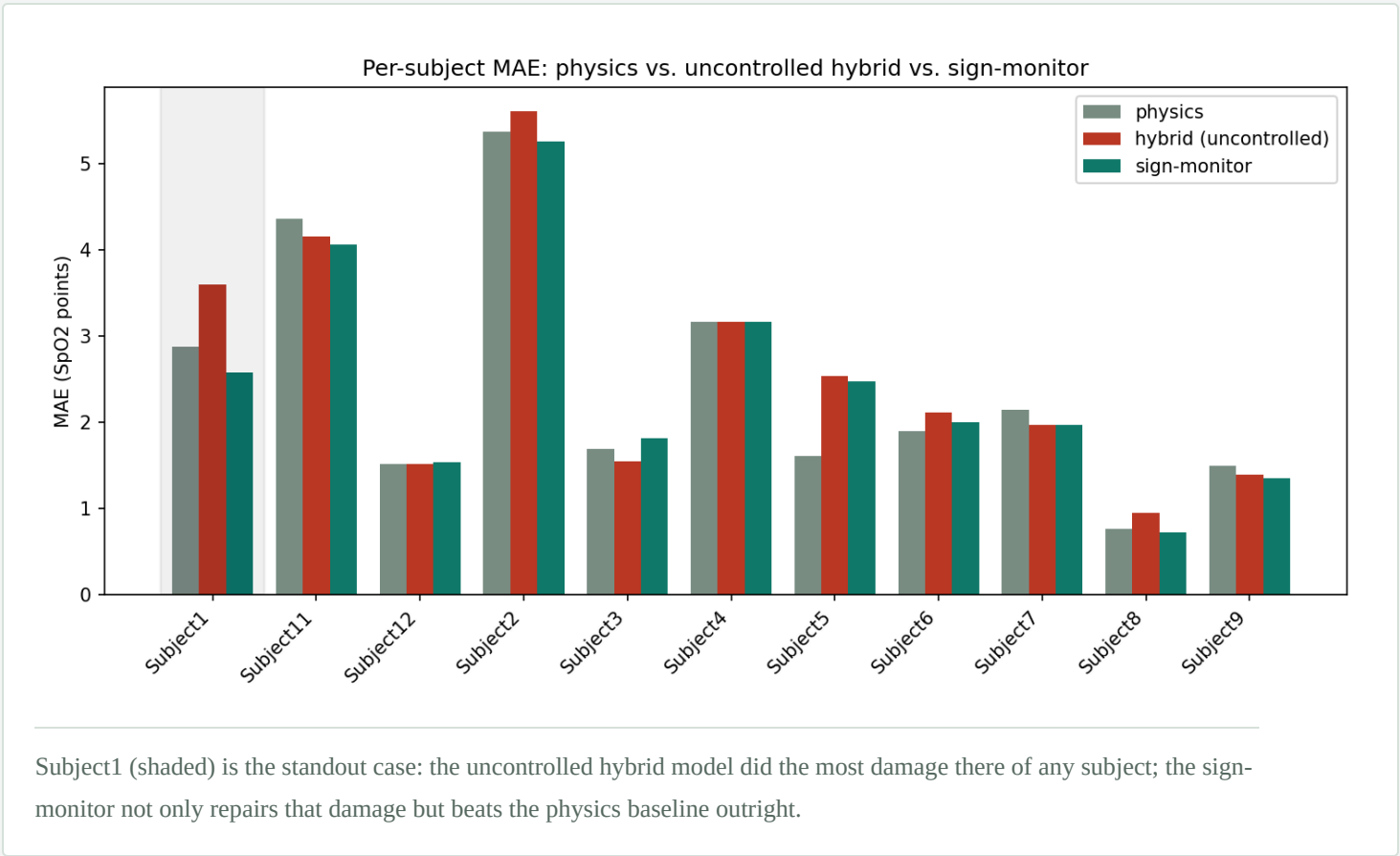


POOLED ERROR, ALL ESTIMATORS

ESTIMATOR	MAE	RMSE
physics	2.287	3.218

ESTIMATOR	MAE	RMSE
hybrid (uncontrolled)	2.383	3.312
hard-monitor	2.278	3.226
sign-monitor	2.291	3.199
soft-monitor	2.258	3.195

By subject-level win count, sign-monitor is the strongest: improves the hybrid on **8/11** subjects and the physics baseline on **7/11**. An unplanned, data-driven finding: the LOSO grid search selected $\beta=0$ on *all 11 folds* — partial attenuation was never preferred over full rejection, so the sign-monitor degenerated into a binary gate (trust unanimous agreement fully, otherwise use physics only).



SUBJECT1, SEPARATELY – THE WORST UNCONTROLLED-HYBRID FAILURE IN THE WHOLE INVESTIGATION			
ESTIMATOR	MAE	RMSE	MAX ABS ERR
physics	2.879	3.714	6.150
hybrid (uncontrolled)	3.596	4.260	7.793
sign-monitor	2.579	3.272	6.150

Worst-subject performance (Subject2, the hardest subject under every estimator) also improves under the sign-monitor: MAE 5.261 vs. the uncontrolled hybrid's 5.609 — the gate helps the tail,

not just the average.

HONEST LIMITS

Not universal: Subject3 and Subject4 do marginally worse under every monitor than under the uncontrolled hybrid or physics alone. The hard monitor's binary on/off gate, despite being the simplest mechanism, is the weakest of the three — it improves the physics baseline on only 5/11 subjects, worse than the sign-based gate's 7/11, because collapsing disagreement to a single scalar threshold throws away information the sign-based gate preserves.

08 • CONCLUSIONS

What this investigation actually established

- **Two real bugs were found and fixed** before any downstream result could be trusted — a BatchNorm freeze leak and a channel-variance-driven Stage-1 training failure, both verified against real run logs, not assumed.
- **The uncontrolled learned correction does not reliably beat the classical physics baseline** on this 11-subject dataset, even after four rounds of capacity/regularization tuning — a genuine negative result, not a bug to keep chasing.
- **Correction magnitude (ρ) predicts outcome magnitude conditional on direction, never direction itself** — the two-fan scatter plot is the clearest evidence in the whole report for this distinction.
- **No single model's own runtime output predicts its correction's direction reliability** — raw logits, embeddings, and MC-dropout uncertainty all sit at or near chance, properly evaluated with LOSO-honest thresholds.
- **Leakage-free cross-model disagreement does predict direction reliability**, and closing the loop on it produces a real, held-out-validated improvement in the actual estimate — not just in a classification metric.
- **The path forward is the sign-agreement or soft-attenuation gate**, not the simpler hard threshold, and not fancier combinations of signals — both underperformed the single best mechanism when properly, honestly evaluated.

loso_fold_predictions.csv · loso_enriched.csv · loso_checkpoints_train3/